

Three identical coins, each of mass m and radius R (thickness neglected), are stacked vertically on a horizontal table, labelled A (bottom), B , C (top). The kinetic friction coefficient and the maximum static friction coefficient are the same everywhere — both between the table and a coin, and between any two coins: μ_k and μ_s respectively, with $\mu_s > \mu_k$. The gravitational field is uniform with strength g . The coins only translate; they never tip.

The table's right edge joins smoothly to a rough inclined plane of angle θ , made of the same material (friction coefficient μ_k).

Assumptions

1. All coins are identical point-like disks of mass m , radius R ; thickness and tipping are ignored.
2. Friction coefficients are the same at every interface: kinetic μ_k , static max μ_s , with $\mu_s > \mu_k$.
3. A coin slides with kinetic friction $\mu_k N$; two coins stay stuck together whenever the required force is below the static limit $\mu_s N$.

Questions.

Q1. Slow push.

1. A horizontal force F is used to push A slowly, so that the three coins just barely begin to slide as one rigid body. Find F .
2. If the force is increased, A begins to slide relative to B while B and C remain stuck together. Find the critical force F_{crit} at which this happens.

Q2. Constant push. Now a constant force $F = 5\mu_k mg$ pushes A ; A slides relative to B , while B and C stay together.

1. Find the acceleration a_B of coin B .
2. From the start of the push until A has completely emerged from beneath B (a relative displacement of $2R$), how much time t elapses?
3. With everything else unchanged, the force is instead $F = 3\mu_k mg$. Qualitatively: does relative sliding occur between A and B ? If not, what motion does the whole system undergo?

Q3. Up the incline. After A has emerged from beneath B , the three coins slide on the horizontal table, and eventually A , B , C rush onto the rough incline together with a common speed v_0 , sliding up a maximum distance s . Over the entire process from the start of the push to the final stop on the incline, the pushing force F does work W_p .

1. Write the relation between the total work W done by all friction forces over the whole process and W_p .
2. Find the magnitude a of the acceleration during the motion up the incline, and write the relation between v_0 and s .